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Next Generation Mathematics: The Universal Number System

1. Abstract

The Universal number system is a newly discovered vast number system with the domain that includes integers, real numbers, and complex numbers in its ecosystem. Also, magnitudes in this ecosystem are precisely controlled for every magnitude including the entire domain of complex numbers contains, and also other magnitudes such as zero, less than zero, infinity, and higher than infinity. Having precise control over such numbers allows us to conduct any operation with these magnitudes such as division by zero or multiplication by infinity, to handle function points precisely that are beyond infinity or below zero and many. This infinitely increased domain range beyond complex numbers helps us in understanding many real life phenomena, science, nature, and the unaversive very accurately in terms of depth and wideness. A few bizarre and interesting things noticed in this system are magnitudes are relative and they change their dimensions in certain circumstances. And this is why indeterminate forms including division by zero have no possible solution by just remaining in the real or complex number system which in turn represents just one dimension.

Using this infinitely high accuracy system, we also understand we are not just approximating the numbers in mathematics but it is found that unknowingly the outcome of equations and formulas are also been approximated for ease or reducing the complexity of the solution. And this leads in yielding a wrong solution in various corners of mathematics. Example: Multiplication of any magnitude by zero is zero which is a correct solution but it is an incorrect solution if we consider a certain level of precision. So, using this system, we can effectively use the approximated solution and handle it without getting into any invalid solution. Also, this system provides the ecosystem to conduct its required operation precisely and without any approximation, which the current system does not provide due to its limitation in the domain range.

So, in this article, its fundamentals, basic principles, and rules are established to support the smooth and balanced operation in the universal number system. Also, some application problems are solved that are generally not possible in the real or complex number system. However, the research on this topic is limited as it's a fresh new discovery, and looking at the current potential, it can be well and easily predicted that a very bizarre understanding and extensive amount of applications are yet to come.

Keywords

Higher Number system, Highly Precise Number System, Multidimension Number system, Newly Discovered Number System, Highest domain range number system, Negative Factorial, Division by Zero, Precise derived Algebraic equation, new precise subtraction rule, new precise division rule, new precise exponential rule, Solved Indeterminate Forms, Zero, Infinity, Infinite Series, Increased Domain Range, Real universe oriented number system, BBUM, Approximation of Equation.

Part I – Theory and Fundamentals

2. Hypothetical Philosophy

2.1 Theory: Multiple Behaviours across multiple dimension

Anything possesses multiple solutions (that may include multiple characters, duality nature, multiple behaviors, ambiguities) because that entity is made up of multiple smaller entities of different behaviors existing in different dimension levels, or in the same dimension but rules, theories, and concepts are defined such that it refers multiple entities as one entity. In other words, if we are getting multiple solutions due our theories are not accurate, rather it is approximating different natural entities as one entity. This also means, if we encounter any multiple solutions, it is a clear indication, there is more in depth research and understanding required.

But in nature, if we do research and build the concept to clear one multiple solution, it holds the chances that there will be a rise of another multiple solution situation in the newly created concept, and multiple solutions will be like a chain reaction that never ends, but the main focus should be, that we are getting the accurate solutions for what specifically we are looking for.

And this theory applies from the infinitely thinnest domain till the infinite largest domain, and also to the smallest thing in this universe till the largest thing in this universe.

2.2 Hypothesis: Existence in Multiple dimension

In the Universal number system, there are numbers in multiple dimensions and similarly, everything in this universe does exist in multi dimensions in the same logic as numbers in the universal number system. And this is because the mathematical framework logic of the universal number system seems to be coinciding with the fundamentals of this universe.

The number in the universal number system does change its dimension if it undergoes a certain specific operation. Example: A number from zero dimension can come in the standard dimension or infinite dimension or in any other dimension but it requires its relevant operation for such transformation. Also for a note, irrespective of what magnitude, the number is representing, its dimension creates a dominant difference if another comparing/relative number belongs to a different dimension.

Real world example:

1. *A human comes to birth, and so his/her life can be said that it has changed its dimension due to certain operations and come to human life from another life. That life may have come from any*

other dimension or the same dimension. Considering human life is in standard dimension, it may have come from zero dimension, infinite dimension, same dimension, or any other dimension.
Also when a human dies, its life entity can go to any dimension. But understanding the universe that it's easy to lose energy than to hold it(in human language, being independent is much easier than being united), the majority are expected to go to a lower dimension. Also, it is easy for energy to go one step higher or lower than going two steps higher or lower, or it's easier to go two steps higher or lower than going three steps higher or lower, and so on. And therefore, it's also expected that most energy will end up in transforming into the nearest dimension rather than far once.

2. *It is very much possible that various sensing organs that human poses are sensing the physical energy that are in one sequence dimension order. For example the light energy, then the sound energy, then the smelling matter, then the tasting matter. All these four forms of energy may be part of consecutive dimension levels without skipping the dimension in between.*
Ex Analogy 1: If light falls in zero cube dimension, then the sound will be in zero square dimension, smelling matter will be in zero dimension and the testing matter will be in standard dimension.
Or this allocation may be slightly different on the universal number line:
Ex: Analogy 2: The light energy is between zero cube and zero square dimension, the sound wave is between zero square and zero dimension, smelling matter is between zero and unity dimension, and the tasting matter is between unity and infinity.
Ex: Analogy 3: The light may be after zero dimension, that is near to zero but not near to unity. Then the sound will be before the unity dimension, near to unity dimension but not near to zero dimension. The smelling matter will be after unity, near to unity but not near to infinity. The tasting matter will be before infinity, near to infinity but not near to unity or infinity square. And considering the behaviors of sound, this analogy is more fitting.
But in which dimension the light energy truly lies is a matter of setting an appropriate initial reference and the rest will follow. Ex: Light energy may fall in zero square dimension and the rest will consecutively follow. Also, light energy may be lying on a higher dimension(eg: infinity cube) and the rest will follow consecutively in lower dimensions in reverse order (lower than the infinity cube). These are all matter of setting the reference as per more understanding of these entities.

Suppose there is some interaction or operation between certain entities, may be clouds(represented by some set of the bulk mathematical equation) which cause lightning. Now when lightning happens, it breaks down or transformed some set of the equation or an individual magnitude, and gets released in many forms such as light, sound, electricity, smoke…. . At the initial stage, there was no light, no sound, no electricity in the clouds but due to some operation, the existing equation in the clouds has changed its dimension causing it to release sound, light, electricity, smoke…….
Mathematically this can happen as: Let's suppose the cloud is existing as the below equation. $500x - 350z - 110k$. *Now due to random uncontrolled physical movement, there is an incoming of negative* $200x$ *in the equation. This will transform into multi dimensional equation as as earlier equation will turn as* $300x + 200x(0) + 350z - 110k$. *Now due to some spontaneous changes, there is a chance to displace these number units and the occurrence of other*

arithmetic operations. And this operation causes spontaneous changes in physical reality and it will put some entity/magnitudes in various dimension such as zero cube dimension, infinity dimension based on which is more likely to occur in nature. And once this equation gets independent they possess their character based on the dominant dimension magnitude in the equation.

And various dimension independent equations can be perceived as different real world entities.

Ex: Analogy 4: Not all words and information we receive can transform into real action, it all depends on a person's existing equation as well as the incoming equation and maybe the critical hit situation also.

In universal numbers, any number can transform into any dimension and there is no such as a basic elementary dimension from where it all starts. Ex: zero divided by zero square will put its magnitude into infinity dimension. So, it's also not possible in the physical world to get an elementary particle that cannot be broken down into other elementary particles. There may be a certain hypothesis that may say zero and one is fundamental numbers from which this all arises. In that case, that may be also true but it is a different perspective of viewing, having different governing rules and fundamentals. And various perspective has their own pros and cons.

2.3 Hypothesis of the physical and non-physical world

The magnitude in the equation represents the energy, the magnitude in the equation is situated at a certain place in the equation that represents the space, and the process the equation goes through represents the time. If the equation doesn't change, there is no time. And so there may be the question itself that why the equation will change. Supporting answers may be many, such as

- *Equation clusters are a mix of multiple various equations, that holds the potential to create an imbalance in the equation system and an imbalance equation attracts a new equation. And such a chain reaction once started, it always continues.*

And equality is a little complex to understand, but at a broad level, the earth is stationary for the physical matters on earth so the earth's center point may represent equality, and the rest of other physical and nonphysical matter around it represents magnitudes in the equation. Since there are many independent magnitudes with various operations, it will spontaneously interact with other magnitudes and yield a different resultant magnitude.

3. Introduction of universal numbers

The universal number system deals with numbers far more accurately than any other number system and it also operates in a far more broader and wider domain than real or complex numbers. This increases the horizon of mathematics both in the depths (smallest magnitude) and wideness (highest magnitude) of mathematics. And this allows us to deal with numbers with

higher accuracy leading to obtain very precise solutions along with dealing in various dimensions of the number system that any other existing number system cannot allow.

A broad understanding of universal numbers can be understood from the number line in figure 1. A number line with positive magnitudes resembles the positive side of the number line and with negative magnitudes resembles the negative side of the number line. And line marked with black colour indicates the domain of the real number line.



Figure 1: Wide view of universal number line. Left side is negative side and right side is positive side of the universal number line.

However, the real number is not accurately defined on the entire black line in figure 1. This is because the real number can find the difference between 1/100 and 1/10000 but it cannot find the difference in ∞/100 and ∞/10000 rather it treats both ∞/100 and ∞/10000 as the same magnitude. Similarly, the Real number can find a difference in 1x100 and 1x10000 but it cannot find the difference in 0x100 and 0x10000 rather it treats both 0x100 and 0x10000 as the same magnitude. In other words, 0 and ∞ are not accurately defined entities in real number line rather it represents only the rough boundary limits. So, figure 2 is the actual real number line in black colour on the universal number line (red colour including black colour) where the real number is accurately defined in the vicinity of unity and not beyond. So basically in figure 2, the red line indicates the magnitude beyond the real number system that is also recognized accurately on the universal number line.



Figure 2: Accurately defined magnitude of Real number line (Black colour) and universal number line (Red colour including black colour line).

The way universal number system covers very wide range of number line, it also covers the very depth of the number line at very high precision which is illustrated in figure 3 and figure 4.



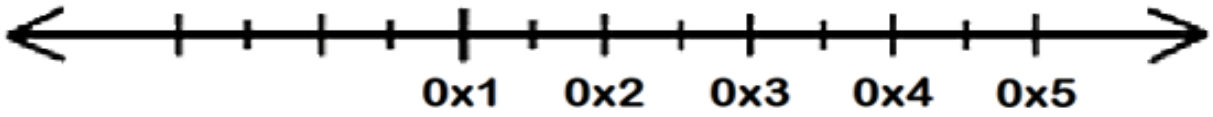


Figure 4: Depth view universal number

4. Dimensions of Number System

The numbers in the vicinity of 0 represent lower dimension magnitude similarly the numbers in the vicinity of 1 represent the standard dimension where the entire real number operates. And in the same order, all numbers in the vicinity to infinity represent the giant dimension. Apart from these, below zero are miniature dimensions, and above infinity are supreme dimensions. Precisely, these dimensions can be also termed with its respective numbers and its powers to resemble dimensions such as zero cube dimension, zero square dimension, zero dimension, unity dimension, infinity dimension, infinity square dimension, infinity cube dimension, and so on. However, the below mentioned dimensions are named separately as it will be used frequently or if required to be used beyond the subject of mathematics.

- Any dimension below Zero Dimension — Miniature Dimension
- Zero — Null Dimension
- Unity — Standard Dimension
- Infinity — Giant Dimension
- Any dimension above Infinity Dimension — Supreme Dimension
- Any dimension Below Standard Dimensions — Lower Dimensions
- Any dimension Above Standard Dimensions — Higher Dimensions
- Zero, Unity & Infinity — Basic Dimensions

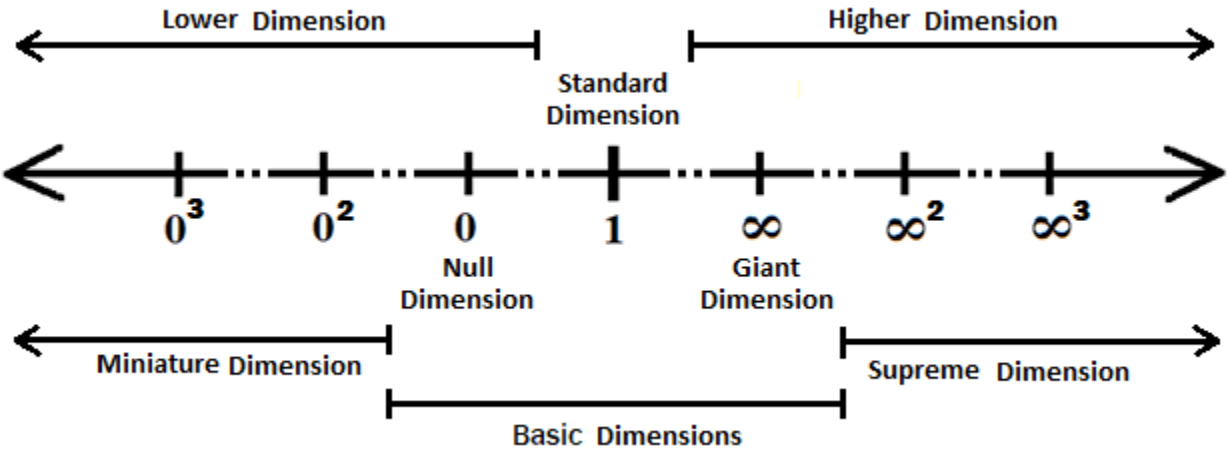


Figure 5: Dimension Terminology in Universal

The basic and Principal Mathematical relation between zero, unity, and infinity is given by

$$0 = 1-1 \tag{1}$$

$$1 = 0 \times \infty \tag{2}$$

$$\infty = \frac{1}{1-1} \tag{3}$$

4.1 Infinity and Zero

Infinity and zero act as fixed values but in relative order and so they are considered as dynamic numbers that do not have any specific static fixed value without any refrance. Rather the value of infinity keeps self incrementing and the value of zero keeps self decreasing at the same rate as infinity. Both inverse infinity and infinity are inverse counter entities to each other. And, if higher infinity is introduced compared to existing infinity, it simply means the rate at which higher infinity is incrementing is more than infinity.

It may be also hinted by the equation infinity minus infinity is one. The difference is one, which is most likely indicating that is how infinity will be. Similarly, zero minus zero is zero square which means infinite zero squares should increment infinite times to become zero. And so this is what the universal number system includes other than the static numbers (Real numbers).

And the relation between zero and infinity is how its dimensions are connected to each other mathematically. These mathematical equations can be also used to construct the mathematical bridge between other higher and lower dimensions. For example, if we want to connect unity to zero square dimension, we have to take zero and unity relation and raise the power on both sides by square.

5. Non-Reduction forms

The magnitude associated with zero or infinity is no further reduced as it indicates in what dimension the magnitudes are existing also if it tried to reduce further, it won't result in a more meaningful equation. However, if such numbers such as zero is replaced with any other equation to solve in better way, then the equation is altogether in a different situation. But while any magnitude is individually present with zero or infinity or any other dimension points, it's better to keep it like that. Below are some of the examples.

1.
$a+0=a+0$

2.
$a-0=a-0$

5.
$a+\infty=a+\infty$

6.
$a-\infty=a-\infty$

3.
$a \times 0 = a \times 0$

4.
$a \div 0 = a \div 0$

7.
$a \times \infty = a \times \infty$

8.
$a \div \infty = a \div \infty$

6. Advanced Rules and Operations of Universal numbers System

The universal number operates at a very high precision magnitude level along with its approach to deal with numbers compare to real number system and these high precision approaches force current mathematical rules and equations to undergo rederivation from basics and formulate the mathematical equation such that equation and its equalities are precisely valid in universal number system too.

And since real numbers or complex numbers are less precise than universal number system, it holds the possibility that some equations are precisely valid both in the complex number system and universal number system, but there may be some equations that are precisely valid only till complex number system and not till universal number system.

This means, there are certain rules, that real number systems including complex number systems are not able to identify those inaccuracies, it is just that these rules are accurate enough to be validated and proved correct at the real or complex number system level.

But in certain circumstances, these inaccurate rules create an invalid solution in the real number system itself, which are handled by creating restrictions in some rules such as some rules or operations are not applicable in few conditions. So, those inaccurate rules are rederived here from basic to match the accuracy at the universal number system and those rules will be known as advanced arithmetic properties of the universal number system that are given below.

6.1 Advanced arithmetic properties of the universal number system

6.1.1 Addition-Subtraction Properties

These all rules are based on valuing every magnitude from every dimension as they hold some value that will create an imbalance in the equation if accidentally these values are ignored.

1.

Real Number $0 + 0 = 0$

Universal Numbers $0 + 0 \neq 0$

Universal Numbers $0 + 0 = 2(0)$ (4)

2.

Real Number $\infty + \infty = \infty$

Universal Numbers $\infty + \infty \neq \infty$

Universal Numbers $\infty + \infty = 2(\infty)$ (5)

3.

Real Number $x - x = 0$

Universal Numbers $x - x \neq 0$

Universal Numbers $x - x = x(0)$ (6)

4.

Real Number $0 - 0 =$ *undefined*

Universal Numbers $0 - 0 = 0^2$ (7)

5.

Real Number $\infty - \infty =$ *undefined*

Universal Numbers $\infty - \infty = 1$ (8)

6.1.2 Multiplication-Division Rule and Operations

6.

Real Number $0 \times 0 = 0$

Universal Numbers $0 \times 0 \neq 0$

Universal Numbers $0 \times 0 = 0^2$ (9)

7.

Real Number $\infty \times \infty = \infty$

Universal Numbers $\infty \times \infty \neq \infty$

Universal Numbers $\infty \times \infty = \infty^2$ (10)

8.

Real Number $0 \div 0 =$ *undefined*

Universal Numbers $0 \div 0 = 1$ (11)

9.

Real Number $\infty \div \infty =$ *undefined*

Universal Numbers $\infty \div \infty = 1$ (12)

10.

Real Number $0 \div 0 = 0$

Universal Numbers $0 \div 0 \neq 0$

Universal Numbers $0 \div 0 = 1$ (13)

11.

Real Number $a \times 0 = 0$

Universal Numbers $a \times 0 \neq 0$

Universal Numbers $a \times 0 = a(0)$ (14)

12.

Real Number $\frac{a}{0} = undefined$

Universal Numbers $\frac{a}{0} = \frac{a}{0}$ (15)

13.

Real Number $0 \times \infty = undefined$

Universal Numbers $0 \times \infty = 1$ (16)

6.1.3 Power Rules and operations

14.

Universal Numbers $$a^0 = 1 + \frac{\ln(a)}{\infty} \quad (17)$$

Universal Numbers $$(-a)^0 = 1 + \frac{\ln(a) + i\pi}{\infty} \quad (18)$$

In the real number system, subtracting any two same numbers yield the same zero but in the universal number system, it yields a different form of zero. Similarly, the division of the same number in the real number system yields the same unity but in the universal number system, it yields a different form of unity. This is because of the basic idea of subtraction between x-x.

So, the general formula for division operation is given by

$$\frac{a}{a} = a^{1-1} = a^0 \quad (19)$$

$$a^0 = 1 + \frac{\ln(a)}{\infty} \quad (20)$$

Prove that RHS is equal to LHS for the equation $a^0 = 1 + \frac{\ln(a)}{\infty}$

$$a^0 = 1 + \frac{\log a}{\infty}$$

Adding infinity to power on both sides of the equation.

$$a^{0\times\infty} = \left(1+\frac{\log a}{\infty}\right)^{\infty}$$

$$a = \left(1+\frac{1}{\infty}\right)^{\infty\log a}$$

$$a = e^{\log a}$$

$$a = a$$

Also, using the 18 formula, one to the power zero can be written by

$$1^0 = 1 + \frac{\ln(1)}{\infty} \tag{21}$$

Adding infinity power on both the side of the equation, it can be written as

$$1 = \left(1 + \frac{\ln(1)}{\infty}\right)^{\infty} \tag{22}$$

Also, natural log unity can be expressed as

$$\ln\left(1\pm\frac{a}{\infty}\right) = \pm\sum_{k=1}^{\infty}\frac{1}{\infty^k}\frac{a^k}{k} \tag{23}$$

6.2 Defined equation of Unity

Unity can be expressed in various forms and also it can be defined as 1=1/1 as there is no contradiction found even raising the unity with infinity on just one side of the equation as in equation (22). In case if there is any noticeable inaccuracy in the real number system, it may be when the exponent power of the unity is a lot greater number than infinity. So, we are in no problem considering this definition as the inaccuracy is very far from the boundary of the currently known universal number system.

15.

Universal Numbers $$1^{\infty} = 1 \tag{24}$$

when $$1 = \frac{1}{1}$$

then

$$1 = 1^{1-1} \Rightarrow 1 = 1^0 \Rightarrow 1^{\infty} = 1 \tag{25}$$

Universal Numbers

$$1^{\infty} = (1+\eta)^{\infty} \tag{26}$$

where η is a negligible magnitude.

6.3 Negligible Magnitude η

There is a difference in the actual magnitude of inverse infinity(zero) and negligible inverse infinity (negligible zero) in the way we consider it. When we consider the inverse infinity as negligible magnitude in an equation, that negligible magnitude is infinity times smaller than any smallest magnitude in the equation and so in general, the inverse infinity will be relatively small. If inverse infinity(zero) is present in the equation as the normal magnitude in the equation (not as a negligible magnitude) and in that equation if we have to add negligible magnitude, then we have to add it such that it is infinitely smaller then the smallest existing number in the equation. So, if zero is existing in the equation, then zero square or smaller than zero square can be considered as negligible magnitude. If zero square is the smallest magnitude the in the equation then the negligible value must be less than zero cube.

Also, when there is infinity or any dimension above this, it should be taken into consideration that infinity represents zero as it also represents as infinity if it moves to another side of the equation. Whether infinity is at a power place or base place, it also signifies zero is present in the equation. For example.

$$x^{\infty} = y \quad \longleftrightarrow \quad x = y^{0} \tag{27}$$

$$\infty \times x = y \quad \longleftrightarrow \quad x = y \times 0 \tag{28}$$

So, if there is an infinity square in the equation, then the negligible magnitude must be at least one dimension lower than the zero square dimension.

Mathematically negligible magnitude is expressed by a smaller Greek letter η (eta).

Where

$$\eta = \varepsilon \times m \quad : \quad \begin{cases} \varepsilon = \textit{smallest dimension in equation} \\ m < 0 \end{cases} \tag{29}$$

To simplify, it can be also written as

$$\eta = 0 \times k \quad : \quad k = \textit{smallest dimension in equation} \tag{30}$$

To make negligible magnitude more aligned to existing concepts of mathematics, it can be expressed as,

$$\eta = a \times \varepsilon \times m \quad : \quad \begin{cases} a \in \mathbb{R},\ a \notin 0 \| \infty \\ \varepsilon \ = \ smallest\ dimension\ in\ equation \\ m = 0 \end{cases} \tag{31}$$

All these components of negligible magnitude make it more specific, accurate, and easy to break down at the accuracy of the universal number system. In this equation, real number and universal number components can be easily separated to deal with it easily and this equation can be also expressed as,

$$\eta = a \times \varepsilon \times 0 \quad : \quad \begin{cases} a \in \mathbb{R},\ a \notin 0 \| \infty \\ \varepsilon \ = \ smallest\ dimension\ in\ equation \end{cases} \tag{32}$$

So this negligible magnitude is the smallest magnitude that can be considered relative to other existing magnitudes in the equation. However, the general negligible magnitude can be expressed from equation (22). As having $0\ln(1)$ in addition to unity creates no difference while raised with infinity. But this cannot be marked as fixed negligible magnitude in all cases in entire mathematics as magnitude relativity can change everything. This equation will be used as negligible magnitude up to a certain extent only due to it showed no effect in equation (22) which is also mathematically proven after equation 20. So,

$$\eta = \frac{\ln(1)}{\infty} \tag{33}$$

$$\eta = 0\ln(1) \quad : \quad a = 1\ ,\ \varepsilon = \ln(1) \tag{34}$$

6.3.1 Limitation of Negligible Magnitude (η)

The result obtained using negligible magnitude cannot be used as a final solution as the outcome of the equation is obtained by adding some mathematical impurity but the pure solution must be obtained without adding negligible magnitude. Solution using plus and minus negligible magnitude $(\pm\eta)$ should be used only to understand if the outcome is about to converge with the outcome value obtained by other formulas. And this method is used where there is no way to get the solution using just a pure form of the equation rather we can solve it by adding negligible impurities.

6.4.1 Indeterminate form, 1^{∞} in the real Number system

One to the power infinity will lead to indeterminate forms in the real number system as the equation itself is ambiguous in the real number system. And the meaning of that equation are many in that system. It treats $n \times \infty = \infty$ which is one of the ambiguities, and due to this

$1^{\infty} = 1^{2\times\infty} = 1^{3\times\infty} = 1^{4\times\infty} .... 1^{n\times\infty}$ and so $1^{\infty} = (1^{\infty})^2 = (1^{\infty})^3 = (1^{\infty})^4 = ... (1^{\infty})^n$ and this leads to conclude $e = e^2 = e^3 = e^4 = ... = e^n$ in the real number system which is an indeterminate form.

6.4.2 Indeterminate form, 1^{∞} in Universal Number system

Now, considering $f(x) = 1^{\infty}$ as a discrete function in the universal number system, it can be expressed and solved as

$$1 = \frac{1}{1}$$

$$1 = 1^{1-1} \tag{35}$$

$$1 = 1^{0}$$

$$1^{\infty} = 1 \tag{36}$$

And considering $f(x) = 1^{\infty}$ as a continuous function in the universal number system, it can be solved using the approaching method by adding a negligible value in the equation. In this equation, zero is already present so the negligible magnitude will be any magnitude from zero square dimension or below. So adding negligible value in one to power infinity, we get

$$1^{\infty} = (1+\eta)^{\infty} \tag{37}$$

Where η is a negligible value.

Or

$$1^{\infty} = (1+a\varepsilon 0)^{\infty} \tag{38}$$

1^{∞} can be also expressed in different forms as below equations

$$(1+a\,\varepsilon\,0)^{\infty} = (1+0)^{a\varepsilon\infty} \tag{39}$$

$$(1+a\,\varepsilon\,0)^{\infty} = 1+\sum_{k=1}^{\infty} \frac{1}{\varepsilon^k}\frac{a^k}{k!} \quad : \quad a \in \mathbb{R} \tag{40}$$

$$(1+a\,\varepsilon\,0)^{\infty} = e^{a\,\varepsilon} \tag{41}$$

So, using equation 40, 1^{∞} can be expressed as equation (42) in a more simpler form.

$$(1+\varepsilon\,0)^{\infty} = 1+\sum_{k=1}^{\infty} \frac{1}{\varepsilon^k}\frac{1}{k!} \tag{42}$$

6.5 Algebraic operations with zero

16.

Real Number $x-y+a-a=x-y$

Universal Numbers $x-y+a-a\neq x-y$ (43)

Universal Numbers $x-y+a-a=x-y+a-a$ (44)

Example:

Let $a=a$

Inserting $-a$ on both sides of the equation

$$a-a=a-a$$

Now if we insert $-5a+5a$ only on one side of the equation

$$(a-a)+(5a-5a)=(a-a)$$

Dividing by $(a-a)$ on both sides of the equation

$$1+5=1$$

$$6=1$$

So, from here we prove that inserting any level of magnitude such as $-5a+5a$ or any other magnitude belonging to any dimension breaks the stability of the equation or forms an imbalance equation. So, any magnitude belonging to any dimension should not be inserted in the equation without balancing it on the other side of the equation.

6.6.1 Insertion property of Equality.

In any equation, any positive or negative magnitude can be inserted independently on both sides of the equation.

Eg: if the given equation is $x-y=x-y$

So, magnitude negative z can be inserted on both sides of the equation

$$x-y-z=x-y-z$$

Remark (1): In any equation, it is invalid to insert positive and negative magnitudes on only one side of the equation.

6.6.2 Elimination and Subtraction

Elimination is the process of reducing or shrinking the existing magnitude into a smaller magnitude in the equation while subtraction reduces the magnitude as well and after a certain magnitude limit, it can change the direction of the magnitude too accordingly. These both terminologies have similar processes except the elimination method only flows in one direction, which is from higher magnitude to lower magnitude, whereas subtraction flows in both directions based on the situation, that is from higher magnitude to lower magnitude and also from higher magnitude to inverse higher magnitude.

1. An example of inverse higher magnitude is,
   let us consider an equation
   $$6 = 6$$
   And if we want to subtract 14 from both the side of the equation,
   $$6 - 14 = 6 - 14$$
   then solution turns out to be
   $$-8 = -8$$
   This magnitude has increased rather than getting reduced.

2. An example of elimination is,
   let us consider an equation
   $$6 = 6$$
   And if we want to subtract 4 from both the side of the equation then
   $$6 - 4 = 6 - 4$$
   then solution turns out to be
   $$2 = 2$$

6.6.3 Super Elimination property of Equality

Any equation remains the same if any independent magnitude is eliminated from both sides of the equation equally and instantly.

Remark for Super Elimination property of Equality

1. Instantly words in the property statement indicate without undergoing any other operation except that particular mathematical operation from the start till the completion of that process.
2. The super eliminator magnitude should be smaller magnitude compared to the magnitude of any side of the equation as this is not subtraction but rather only elimination.
3. The Super eliminator magnitude in the equation will be marked with "right aero bar" on top of the magnitude and it is illustrated as $\vec{x}$.

Note: Super eliminator magnitude is not used for any other operation such as multiplying, dividing, raising powers, etc rather it is only used to reduce the magnitude of the equation, and it enters as independent magnitude with the positive or negative sign in the equation.

Elimination Process

If magnitude x needs to be eliminated from the equation, then super eliminator magnitude $\vec{x}$ (x with right aero bar) will get inserted in both sides of the equation with its positive or negative sign and it will totally eliminate its countersign magnitude from both sides of the equation.

If the given equation is

$$x - y = x - y$$

$$x - y - \vec{x} = x - y - \vec{x}$$

$$-y + \overrightarrow{x - x} = -y + \overrightarrow{x - x}$$

$$-y = -y$$

Remark (2): In any equation, it is invalid to insert positive and negative super eliminator magnitude on only one side of the equation.

6.6.4 Advancement of elimination property of equality

1. The elimination in the sub bracket is possible if the sub bracket can be taken out in the outermost bracket and elements of that sub brackets are exactly the same on both sides of the sub bracket in the equation.

   Process

   Let the equation be as

   $\left[(b - a + m \times n)(c \times d)\right] = \left[(l - k)(c \times d)\right]$

   taking $(c \times d)$ common on both sides of the equation

   $\left[(b - a + m \times n)\right](c \times d) = \left[(l - k)\right](c \times d)$

   And if we want to eliminate from $(c \times d)$ the bracket

   then,

   $\left[(b - a + m \times n)\right](c \times d - \vec{d}) = \left[(l - k)\right](c \times d - \vec{d})$

2. If brackets are inside any other bracket or raised with any powers then all the components of the main bracket that represent the outermost bracket must have exactly the same elements in the same order on the other side of the equation too. And powers must also be the same if brackets are raised with any powers.

3. Only one side of the magnitude, can be eliminated along with the elimination of the super eliminator magnitude of that side and leaving the super eliminator magnitude present on the other side of the equation, condition to
    - Any operations are restricted if it changes the magnitude of that remaining super eliminator magnitude. These restricted operations are such as multiplying the entire equation with any magnitude, increasing the exponential power of the entire equation, etc.

    Example: If the given equation is

$$x - y = z - y$$

    x need to be eliminated from the equation

$$x - y - \vec{x} = z - y - \vec{x}$$

$$-y + \overrightarrow{x - x} = z - y - \vec{x}$$

$$-y = z - y - \vec{x}$$

    - Regardless of having a super eliminator already in the equation, another super eliminator can be inserted on both sides of the equation and this new super eliminator can change the magnitude of the existing super eliminator magnitude. And the arithmetic operation between super eliminator will be by the rules of the Super Elimination property of Equality.

Remark (3): In any equation, it is invalid to insert positive and negative super eliminator magnitude on only one side of the equation.

6.6.5 Magnitude Reversal property of Equality.

Any equation remains the same if any independent magnitude or total magnitude of the equation is reversed up to any magnitude on both sides of the equation equally and instantly.

Remarks for Magnitude Reversal property of Equality

1. Magnitude reversal operation is exactly the same as subtraction operation in a real number system but with some conditions while using it.
2. Independent magnitude means a magnitude separated from another magnitude by a positive or negative sign and that positive or negative sign is not in any bracket.
3. Instantly word indicates without undergoing any other operation except that particular mathematical operation from the start till the completion of that process.
4. Reversal magnitude in the equation will be marked with a "bar" on top of the magnitude and it is illustrated as $\bar{x}$.

Remark (4): In any equation, it is invalid to insert positive and negative reversal magnitude on only one side of the equation.

Note: Reversal magnitude is not used for any other operation such as multiplying, dividing, raising powers, etc rather it is only used to reduce or reverse the magnitude of the equation, and it enters as independent magnitude with the positive or negative sign in the equation.

1 If x amount of magnitude needs to be reversed in the equation, then reversal magnitude $\bar{x}$ (x bar) will get inserted in both sides of the equation with its positive or negative sign and it will totally reverse the relevant magnitude from both sides of the equation.

If the given equation is

$$x+y=x+y$$
$$x+y-\bar{x}=x+y-\bar{x}$$
$$y+\overline{x-x}-=y+\overline{x-x}$$
$$y=y$$

6.6.6 Advancement of Magnitude Reversal property of Equality.

1. The magnitude reversal in the sub bracket is possible if the sub bracket can be taken out in the outermost bracket and elements of that sub brackets are exactly the same on both sides of the sub bracket in the equation.

Process

Let the equation be as

a. $\left[(b-a+m\times n)(c\times d)\right]=\left[(l-k)(c\times d)\right]$

taking $(c\times d)$ common on both sides of the equation

b. $\left[(b-a+m\times n)\right](c\times d)=\left[(l-k)\right](c\times d)$

And if we want to reverse magnitude d from $(c\times d)$ bracket then,

c. $\left[(b-a+m\times n)\right](c\times d-\vec{d})=\left[(l-k)\right](c\times d-\vec{d})$

2. If brackets are inside any other bracket or raised with any powers then all the components of the main bracket that represent the outermost bracket must have exactly the same elements in the same order on the other side of the equation too. And powers must also be the same if brackets are raised with any powers.

3. Only one side of the magnitude, can be eliminated along with the elimination of the super eliminator magnitude of that side and leaving the super eliminator magnitude present on the other side of the equation, condition to

- Any operations are restricted if it changes the magnitude of that remaining super eliminator magnitude. These restricted operations are such as multiplying the entire equation with any magnitude, increasing the exponential power of the entire equation, etc.

  Example: Let be the equation as

  $$x - y = z - y$$

  x need to be eliminated from the equation

  $$x - y - \vec{x} = z - y - \vec{x}$$

  $$-y + \overrightarrow{x - x} = z - y - \vec{x}$$

  $$-y = z - y - \vec{x}$$

- Regardless of having a super eliminator already in the equation, another super eliminator can be inserted on both sides of the equation and this new super eliminator can change the magnitude of the existing super eliminator magnitude. And the arithmetic operation between super eliminator will be by the rules of the Super Elimination property of Equality.

Remark (5): In any equation, it is invalid to insert positive and negative reversal magnitude on only one side of the equation.

7. The critical condition for $f(y) = x - x$ subtraction.

The operation $f(y) = x - x$ is more than what we know from the real number system, it is a reference resetter. It means it will bring any magnitude to the base reference point, but the base reference for every number is relative in a universal number system and not a fixed point like zero in the real number system. And this reference is one dimension cycle backward in the direction of magnitude countdown for every number. Mathematically this is given by $x - x = \pm x(0)$. But it is unknown whether it should be positive or negative zero, which forces us to have 2 solutions in this case as illustrated in figure 6. However, this should not have any effect to the real numbers as the real number system does not count any zero dimension numbers, and nor real number system considers negative zero to be different from positive zero.

So

$$x - x = -x + x = \pm x(0)$$

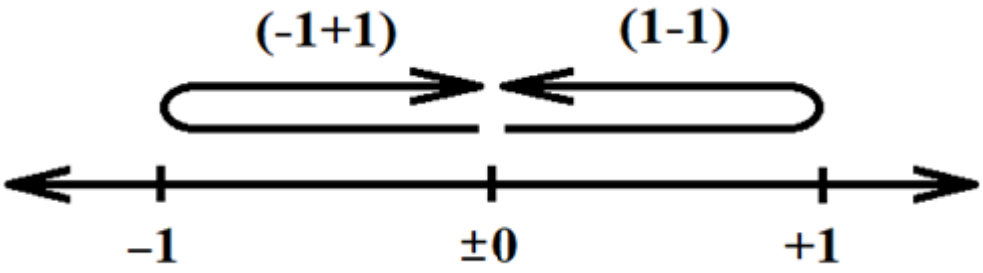


Figure 6: Positive and negative zero as multiple

7.1 Single solution for $f(y) = x - x$

As $f(y) = x - x$ yields more than one solution and considering the theory of multiple behaviors, we conclude that we are referring to multiple entities as one entity. So, now applying all the probability on this equation to distinguish this equation in multiple ways or at least 2 different ways, it has only one set of combinations. and that combination $x - x$ can be written as

1. $+x - x$
2. $-x + x$

And since this is the only combination existing, this is the only way to consider a unique solution. It means both combinations have different meanings in the universal number system. And knowing there is only one set of possibilities, it is the only solution leading to the creation axiom 1. And before we totally accept this solution, the only validation that must be done for such an axiom is, it is not breaking the equality rule (RHS=LHS) of the equation.

Axiom 1: The order of reading the positive operation before the negative operation creates a different meaning than reading the negative operation before the positive operation.

The reading order may not seem different if we look at the equation but if we see it on the number line and conduct the step by step operation, both operation looks clearly different as illustrated in figure 7. So these 2 equations, though looking the same due to the general understanding that $+1-1$ and $-1+1$ are the same but having no more possibility to be illustrated in a different mathematical logical arrangement, we are forced to accept positive and negative zero in this order.

On the number line, it is clear that negative zero and positive zero are not the same and these two points are entirely at different positions on the number line space and the process it undergoes, is also clearly distinctive and is expressed below in equation (45) and (46) for the value of x as one. And for the other value of x, x can be taken common out on both sides of the equation and now it can be again expressed as the same equation and illustrated in figure 7.

Positive zero $+0 = +1-1$ (45)

Negative zero $-0 = -1+1$ (46)

The same thing can be expressed on the number line as in figure 7.

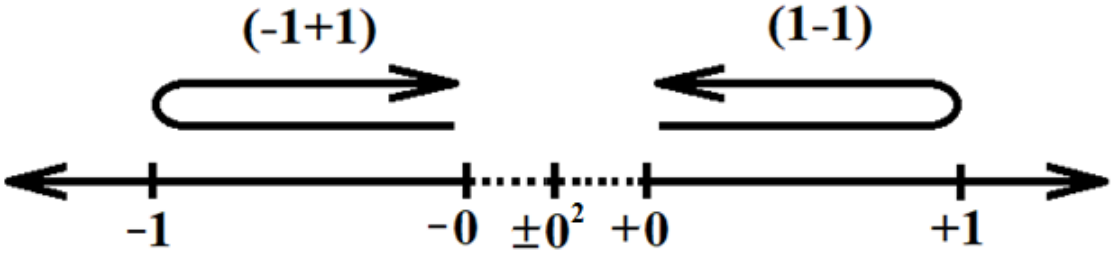


Figure 7: Positive and negative zero as distinguishing solution

But the biggest challenge is about which number (positive or negative) should be read first and on what biases. The only order rule that can support this partially is the BODMAS rule. But the order between subtraction and addition is not strict and can be interchanged in the real number system. So, for the universal number system, these orders must be followed such that all additions are done before any subtraction. Another challenge is, the equation that is written cannot be considered to be existing in its correct order. So we have to follow all addition operations before any subtraction operation, as a rule, to read the magnitude in the correct order. But from this approach also, a single solution cannot be extracted such that it is mathematically proven from other methods. It is in the situation that any one solution from both solutions is in a position to be accepted as a unique permanent solution without breaking mathematical rules in the equation. The solution for equations (45) and (46) can be mapped with any of two solutions that are positive zero and negative zero. But once it is mapped, it needs to be mapped permanently with the same solution to avoid any unknown issues. Using this approach, the solution is distinctively defined in the next section 7.2.

7.2 Defined Single Solution of x-x

Now knowing the logic of the zero in the universal number system, when zero appears in one side of the equation, it also appears on another side of the equation.

Eg: $12-5 = 7\pm5(0)$

$$7\pm5(0) = 7\pm5(0)$$

And the fact to consider is if zero appears on one side of the equation as positive zero x (positive zero x means, positive zero with x magnitude), on the other side also it will appear as positive zero x considering the equation is mathematically correct. And in case if it appears as negative zero x on one side, it will also appear as negative zero x on the other side of the equation too. And now considering the next step, if it is positive zero x on both sides of the equation, it can be

changed to negative zero x by multiplying the entire equation by negative one. And if there is negative zero x, it can be changed to positive zero x by multiplying the entire equation by a negative one without breaking the stability or equality of the equation. So, it does not matter whether the x-x leads to positive zero x of negative zero x. The only thing that has to be taken care of is, everywhere it is yielding zero with the same sign for x-x operation, and the equation will always remain balanced. So now, in this case, we are in the situation to create the decision for the solution of x-x. Should it be positive zero or negative zero? So, for simplicity reasons, the standardized solution will be positive zero as the only solution for the operation of x-x.

Let us consider $a-b=c$ where $|a|>|b|$

We can write

$$a-b=a-b$$

Inserting $-b+b$ on both sides of the equation

$$a-b-b+b=a-b-b+b$$

$$c-b+b=c-b+b$$

If $x-x=+x(0)$

$$c+b(0)=c+b(0) \tag{47}$$

Using the rule of super elimination,

$$c=c \tag{48}$$

Or

$$b(0)=b(0) \tag{49}$$

So, there is no breaking of any stability of the equation.

If $$x-x=-x(0)$$

$$c-b(0)=c-b(0) \tag{50}$$

Using the rule of super elimination,

$$c=c \tag{51}$$

$$-b(0)=-b(0) \tag{52}$$

Multiplying by negative one with the entire equation

$$b(0) = b(0) \tag{53}$$

So, this equation is also not breaking any stability of the equation. Also, comparing equation (48) with equation (51) and equation (49) with equation (53), it is clear that both will result in the same solution.

Both signs do work, but for standard purposes, the solution for x-x is positive zero x.

7.3 Standard solution for x-x

Now we are aware that any naturally occurring any subtraction between the same magnitude will yield a positive zero time that magnitude, we also have to handle the 2 sets of possibility that was occurring with is mentioned in section 7.1.

So, 2 sets of possibilities are +x-x and -x+x and so the expansion of positive zero and negative zero is defined as

Positive zero $+0 = (+1-1)$ (54)

Negative zero $-0 = (-1+1)$ (55)

So, generally in any equation subtraction between the same magnitude should yield a solution as equation (54) and not as equation (55). Equation (55) will be expressed only if the equation is multiplied by a negative one. In other words, if a negative one goes inside the bracket then only the RHS side of equation A will turn into the RHS side of equation (55).

This is also required to be defined to maintain equality in mathematics as using this in random order may result in losing the equality stability of the mathematical equation at the universal number system level.

1. x-x operation leads to one dimension cycle backward in the direction of the magnitude countdown for every number
2. $$x - x = +x(0)$$
3. Since the standard solution for x-x is considered positive, in every operation the same standard must be followed, or else it will lead to wrong solutions. In other words, using negative zero for x-x operation is restricted.

7.4 Arithmetic operation order

The operation order by BODMAS rule is not very critical in real number systems such as the parentheses can be neglected if it only contains addition and subtraction magnitudes inside parenthesis.

Case 1: Boundary condition

Zero minus zero is given by

$$0-0=0^2$$

This can be also written as

$$(1-1)-(1-1)=0^2 \tag{56}$$

And it can be expressed back as

$$(0)-(0)=0^2$$

$$0-0=0^2$$

$$0(1-1)=0^2$$

$$0(0)=0^2$$

$$0^2=0^2 \tag{57}$$

But, if we dissolve the bracket in equation (56)

$$(1-1)-(1-1)=0^2$$

$$1-1-1+1=0^2 \tag{58}$$

Then we can solve equation (58) as previously solved ending up to equation (57), or we can solve it as

$$+2-2\neq 0^2$$

$$2(0)\neq 0^2 \tag{59}$$

Here, RHS is not equal to LHS and this is due to not solving the bracket first but rather dissolving it and conducting the mathematical operation in random order.

7.5 Division by same magnitude

From equation (20), we conclude that division by the same unity is not always unity but rather different magnitudes of unity. These different forms of unity will yield different numbers in a real number system if it is raised by infinity. But the formulation is not easy to handle unless some computational power is used. So in order to minimize the complexity, the answer can be considered at unity and not including other ignorable factors but things need to be taken into consideration that the solution will become unstable if that solution undergoes multiplication with any magnitude that is in infinity square dimension or higher and also if solution power gets raised by infinity.

Case 1. Power of solution when raised by infinity.

Let us suppose $a/a=1$, in this case, the solution will remain unity even if power is raised by infinity. But actually, it should not, as

$$a^0 = 1 + \frac{\ln(a)}{\infty} \tag{60}$$

When raised by infinity,

$$\left(a^0\right)^\infty = \left(1 + \frac{\ln(a)}{\infty}\right)^\infty \tag{61}$$

$$\left(a^0\right)^\infty = \left(1 + \frac{1}{\infty}\right)^{\infty \ln(a)}$$

$$a^{0\times\infty} = \left(1 + \frac{1}{\infty}\right)^{\infty \ln(a)}$$

$$a = e^{\ln(a)}$$

$$a = a \tag{62}$$

And this solution is more accurate at a higher domain range.

Case 2:

If we multiply unity by infinity square, we only get infinity square. But if we multiply the infinity square by the actual solution then the magnitude is much higher than the infinity square.

$$a^0 = 1 + \frac{\ln(a)}{\infty} \tag{63}$$

Multiplying by infinity square or higher magnitude.

$$a^0 \times \infty^2 = \left(1 + \frac{\ln(a)}{\infty}\right) \times \infty^2 \tag{64}$$

$$a^0 \times \infty^2 = \infty^2 + \infty^2 \frac{\ln(a)}{\infty}$$

$$a^0 \times \infty^2 = \infty^2 + \infty^0 \infty \ln(a) \tag{65}$$

And here we see that the magnitude in the solution is much higher than the infinity square.

Part II - Application

The universal number system is not just a specific method or procedure in mathematics that has a specific set of applications. It is the system or the environment that includes all the numbers in it. The discovery of this system has increased the horizon of current mathematics that can end up with infinite advantages or applications based on exploration or conducted research. Since this is at a very early stage, and an unexplored environment, only a few things are discovered that add advantages and applications, and also fulfilling the gaps of other existing numbers systems. As per this research, the general areas of advantages found are

- If the problem is yielding or expected to yield the wrong solution while using the rules at the real number domain then the universal number system can be used.
- If certain operations are restricted in the real number domain, then this system can be used. Eg, division by zero.
- To remain accurate in conveying the question and avoided the ambiguity of the question. Eg: 1-1+1-1+1…. = ?
- If the problem itself is expected to operate beyond the real or complex number domain.
- The universal number system has its own universe where it has lots of applications and understanding that connects the real universe.

7. Accuracy Enhancement of calculus formulation

For deriving calculus formulas, precision is maintained up to a certain extent and not totally. While deriving formulation for calculus, after certain steps the accuracy at the universal number system level is compromised due to neglecting the zero and other lower dimension magnitudes. However, that accuracy is well maintained to remain validated till real/complex number system and so the formulas. But, those formulas are not accurate, but rather with ambiguous terms. So it inherits all the limitations too that exist in other parts of the real number system. For example, if in algebra, division by zero is restricted as it puts the equation in imbalanced form, the same thing will get transferred to calculus too as algebra is its base.

Another aspect of calculus is approaching zero value which has no clear algebraic numerical value from the real number perspective nor empirically it indicates any specific zero in universal numbers, as there are infinitely different zeros on the universal number line but for the real number domain, all those different zeros are same so it doesn't matter for the real number system. It only indicates it is approaching to zero, and it is a very negligible value but the actual value at

the precision of the universal number level is unknown. So, approaching zero will be kept as an unknown or a negligible variable value.

So, here we will evaluate the example formulas that are accurate to their respective domain.

7.1 Derivative formula for constant

Let

$$y = c + 0x$$

$0x$ is the idea of an independent variable that is required for multiple reasons. One of which is the equation will produce a null value (null value is not yet mathematically defined) on only one side of the equation. Also, the operation that this equation is about to undergo is based on the nature of the variable which is zero in this case. And also it gives a clear understanding in this mathematical equation that the idea of an independent variable is an important requirement for this calculus equation. So, being the independent variable is an important variable in this equation, we are expecting this equation to give information about the independent variable where in the case of constant, the value of the independent variable is zero as per the real number system.

$$y + \Delta y = c + (x + \Delta x)0$$

We know that, $y = c$

$$c + x0 + \Delta y = c + (x + \Delta x)0$$

$$c + x0 + \Delta y = c + x0 + \Delta x0$$

$$\Delta y = \Delta x0$$

$$\frac{\Delta y}{\Delta x} = 0 \tag{66}$$

Sam equation can be also expressed for real numbers as there is no further reduction.

$$\frac{\Delta y}{\Delta x} = 0 \tag{67}$$

So, Derivative formula for constant is

$$\frac{\Delta y}{\Delta x} = 0 \ : \in \mathbb{U} \tag{68}$$

$$\frac{\Delta y}{\Delta x} = 0 \qquad : \in \mathbb{R} \tag{69}$$

Where Δx is some unknown negligible value and not an absolute zero (1-1=0).

7.2 Derivative for variable x

$$y = x$$

$$y + \Delta y = x + \Delta x$$

We know that, $y = x$

$$x + \Delta y = x + \Delta x$$

$$\Delta y = \Delta x$$

$$\frac{\Delta y}{\Delta x} = 1 \tag{70}$$

So, the derivative formula for x is

$$\frac{\Delta y}{\Delta x} = 1 \qquad : \in \mathbb{U} \tag{71}$$

$$\frac{\Delta y}{\Delta x} = 1 \qquad : \in \mathbb{R} \tag{72}$$

Where h is negligible value and not an absolute zero (1-1=0)

7.3 Derivative for variable x^2

$$y = x^2$$

$$y + \Delta y = (x + \Delta x)^2$$

We know that, $y = x^2$

$$x^2 + \Delta y = (x + \Delta x)^2$$

$$x^2 + \Delta y = x^2 + 2x\Delta x + \Delta x^2$$

$$\Delta y = 2x\Delta x + \Delta x^2$$

$$\Delta y = \Delta x(2x + \Delta x)$$

$$\frac{\Delta y}{\Delta x} = 2x + \Delta x \tag{73}$$

For the real number system, this can be further reduced below, where Δx is some unknown negligible value and not an absolute zero (1-1=0)

$$\frac{\Delta y}{\Delta x} = 2x \tag{74}$$

So, the derivative formula for x^2 is

$$\frac{\Delta y}{\Delta x} = 2x + \Delta x \qquad : \in \mathbb{U} \tag{75}$$

$$\frac{\Delta y}{\Delta x} = 2x \qquad : \in \mathbb{R} \tag{76}$$

So, as there is an extra term in equation (75) compared to equation (76), it can be said that both have chances to yield a different solution. The way in equation (70), magnitude one is an outcome of division by Δx to entire equation, if a similar operation such as division by Δx or subtraction by Δx occurs in the problem then it is expected to yield a wrong solution. But since it is an ignorable small magnitude, the probability of getting into the wrong solution is very rare.

Hence, many calculus formulas may be correct as we derived for constant and variable x, but there will be also formulas like we derived for x^2 that are accurate only till the complex number domain and not accurate till the universal number system level. And therefore it is required to derive all the calculus formulas again as per the universal number system's equality accuracy.

8. Ambiguities and fallacy manipulation of mathematical terms.

8.1 Ambiguities with infinities

Infinities can be of many infinite forms in the universal number system. But all those various forms of infinities are taken the same in the real number system. So, it has to be well taken care of, to mathematically define the exact infinity in the problem as various types of infinity will yield various results.

Below equation (51) is ambiguous for the universal number level that needs to be more precise while making the question or a statement. This equation will yield various results based on what magnitude of infinity is taken into consideration. Below are examples of various cases that act as the same cases in the real number system but it is different in the universal number system.

Solve

$$x = 1-1+1-1+1... \tag{77}$$

Case 1:

When Infinity is even and infinity is absolute infinity.

$$\sum_{x=0}^{\infty}(-1)^x = +1-1+1-1+1-1...$$

$$\sum_{x=0}^{\infty}(-1)^x = (1-1)+(1-1)+(1-1)...$$

Since 1 is repeated infinite times, then $(1-1)$ will be repeated $\infty/2$ times. So,

$$\sum_{x=0}^{\infty}(-1)^x = (1-1)\,\frac{\infty}{2}$$

As $1-1=0$, we get

$$\sum_{x=0}^{\infty}(-1)^x = (0)\,\frac{\infty}{2}$$

As $0\times\infty = 1$, we get

$$\sum_{x=0}^{\infty}(-1)^x = \frac{1}{2}$$

Case 2: When Infinity is 2∞,

$$\sum_{x=0}^{2\infty}(-1)^x = +1-1+1-1+1-1...$$

$$\sum_{x=0}^{2\infty}(-1)^x = +1-1+1-1+1-1...$$

Since 1 is repeated 2∞ times, then $(1-1)$ will be repeated ∞ times. So,

$$\sum_{x=0}^{2\infty}(-1)^x = (1-1)\,\infty$$

As $1-1=0$, we get

$$\sum_{x=0}^{2\infty}(-1)^x = (0)\,\infty$$

As $0\times\infty=1$, we get

$$\sum_{x=0}^{2\infty}(-1)^x=1$$

8.2 Fallacy manipulation

The infinite series deal with infinity but without an understanding of infinity itself by remaining in the real number domain. Moreover, the unity dimension numbers are underestimated and blindly added or neglected if infinity is present in the equation. And these unity dimension numbers that are added in the equation without balancing the equation put the equation in the possibility that it can yield an unequal result based on what operation its undergone.

The sum of infinite series of natural numbers, its general formulation is given in equation (78), and it must yield the solution as equation (79) but in many journals and mathematical books, its solution is considered as given in equation (80) with the proper set of proofs. And now comparing the solution in equations (78) and (80), it is clear that both solutions are totally different. This implies, at least one solution is a fallacy out of these two solutions. So, while proving the formulation for (78), there are no errors encountered but while proving the solution as equation (80), in every proof the equation is not balanced till the accuracy of universal numbers system level which may be resulting in a different solution.

$$1+2+3+4+.....\ n=\frac{n(n+1)}{2} \tag{78}$$

$$1+2+3+4+.....\ \infty=\frac{\infty(\infty+1)}{2}$$

$$1+2+3+4+.....\ \infty=\frac{\infty^2+\infty}{2} \tag{79}$$

This means the sum is less than ∞^2 but still in ∞^2 dimension and not in ∞ dimension.

But there exists, many other derivations that prove the solution of this series is negative one by twelve as given in equation (54).

$$1+2+3+4+.....\ n=-\frac{1}{12}\quad :\quad n=\infty \tag{80}$$

This difference in solution is due to not having a clear understanding of all mathematical concepts and magnitudes. And this leads to conduct an imbalanced mathematical operation that results in

the wrong solution. As in many places, one is added to infinity and still, it is treated as the same infinity, and the equation is still considered as a balanced equation. And if the solution in equation (54) is true then, it is only due to the question itself have changed in the middle of the operation without giving any hint that the question has changed.

8.3 **Fallacy manipulation of infinite series.**

It must be well taken into consideration that expressing infinity may generally look the same in series form but it may be actually different which will yield a different result. For example, equation (81), (82), and (83) will look exactly the same if it is expressed in the simple form that in equation (84).

$$\sum_{x=0}^{\infty}(-1)^{x} = +1-1+1-1+1-1... \tag{81}$$

$$(1-1)+\sum_{x=0}^{\infty}(-1)^{x} = +1-1+1-1+1-1... \tag{82}$$

$$\sum_{x=0}^{2\infty}(-1)^{x} = +1-1+1-1+1-1... \tag{83}$$

$$x = +1-1+1-1+1-1... \tag{84}$$

As mentioned in the article "Universal Numbers -The Universe of Eternity", indexing scale plays an important role in infinite series. However, it is revised and defined in this section with additional details. This indexing scale is first defined for infinite series as it will be better understood with infinite series.

Indexing scale of series: The overall closeness a series equation is trying to get to its final value with each additional term in the infinite series.

Indexing Unit of series: The total number of units in the equation of the series.

So, any infinite series is said to be the same if the series has the same indexing unit and indexing scale. So, if we add zero to an infinite equation, it may be considered approximately the same as the earlier equation but it should be noted that it is no more the same as the previous equation at the accuracy of universal number system level as its indexing units or indexing scale are now different. And such equations hold the potential to give inaccurate solutions in certain cases if it is treated as the same equation.

9. Factorial of negative integers

The concept of the negative factorial is as same as the positive factorial and the equation for the positive factorial is used to derive the equation for the negative factorial using an empirical method in the research article "Universal Numbers - The Universe of Eternity". And it is concluded that the equation for negative factorial is almost similar but in the inverse order with some additional aspects.

The basic equation for negative factorial is given as

$$(-n)! = \frac{1}{(-n+\bar{1})(-n+\bar{2})(-n+\bar{3})....(-2)(-1)(0)}$$

Remark (6): Reversal magnitude is used to indicate the elimination of those specific magnitudes completely within the bracket as this formula is constructed with the subtraction logic in the real number system. To indicate the previous number in the real number system, $(-n+1)$ or $(n-1)$ mathematical expressions are used, unfortunately, it is not the same in the universal number system rather the reversal magnitudes are used to indicate the previous number such as $(-n+\bar{1})$ or $(n-\bar{1})$.

Example:

$$\frac{1}{(-6)(-5)(-4)(-3)(-2)(-1)(0)} = \frac{1}{(-7+\bar{1})(-7+\bar{2})(-7+\bar{3})....(-2)(-1)(0)}$$

Though both equations have certain similarities such as both are multiplication of all consecutive incrementing numbers, there are other differences that create the differences in their properties. Where the differences are

- The incrementing for positive factorial n starts at 1 and ends at n but for negative factorial, it starts at 0 and ends at – (n-1).
- Also, the incremental values are in the numerator for positive factorial whereas the incremental values are in the denominator for negative factorial.

Example:

$$(7)! = (7)(6)(5)(4)(3)(2)(1)$$

$$(-7)! = \frac{1}{(-6)(-5)(-4)(-3)(-2)(-1)(-0)}$$

And this difference creates differences in their properties and the most commonly used property are as below.

Factorial Properties

Positive factorial

$$(n)! = n(n-\bar{1})!$$

Example

$$(7)! = 7(6)!$$

Negative factorial

$$(-n)! = \frac{(-n+\bar{1})!}{(-n+\bar{1})} \tag{85}$$

Example:

$$(-7)! = \frac{(-6)!}{(-6)}$$

Similarly,

$$(-n)! = \frac{(-n+\bar{2})!}{(-n+\bar{1})(-n+\bar{2})} \tag{86}$$

$$(-7)! = \frac{(-5)!}{(-6)(-5)}$$

So, the generalized form for negative factorial is given by

$$(-n)! = \pm\frac{\infty}{(n-\bar{1})!} \tag{87}$$

Also, there is the possibility that zero in the denominator may be positive or negative.

So, let us substitute zero in the formula $(n)! = n(n-\bar{1})!$

$$(0)! = 0(0-\bar{1})!$$

$$1 = 0(1-1-\bar{1})!$$

$$1 = 0(-1)!$$

$$\frac{1}{0} = (-1)!$$

$$\infty = (-1)!$$

And so, negative one factorial is positive infinity which means positive zero in the denominator.

And, negative factorial pattern example will be

$$(-7)! = \frac{1}{(-6)(-5)(-4)(-3)(-2)(-1)(0)}$$

Or

$$(-7)! = \frac{1}{(-7+\bar{1})(-7+\bar{2})(-7+\bar{3})....(-2)(-1)(0)}$$

Therefore, the formula for such a repeating pattern can be expressed as

$$(-n)! = \pm\frac{\infty}{(n-\bar{1})!}$$

$$(-n)! = (-1)^{n+1}\frac{\infty}{(n-\bar{1})!} \tag{88}$$

Example: Evaluate $(-6)!$

$$(-n)! = (-1)^{n+1}\frac{\infty}{(n-\bar{1})!}$$

$$(-6)! = (-1)^{6+1}\frac{\infty}{(6-\bar{1})!}$$

$$(-6)! = (-1)^{7}\frac{\infty}{(5)!}$$

$$(-6)! = (-1)\frac{\infty}{(5)(4)(3)(2)(1)}$$

$$(-6)! = \frac{-\infty}{120}$$

9.1 Proof for general negative factorial equation

This equation can be proved with the help of mathematical indication.

Step 1

Bases cases for $n = 1$

$$(-n)! = (-1)^{n+1} \frac{\infty}{(n-\bar{1})!}$$

$$(-1)! = (-1)^{1+1} \frac{\infty}{(1-\bar{1})!}$$

RHS will be

$$(-1)! = \frac{\infty}{(0)!} = \frac{\infty}{1} = \infty$$

$$(-1)! = \infty$$

To prove LHS, let us consider

$$n(n-\bar{1})! = n!$$

So, when $n = 0$

$$0(0-\bar{1})! = 0!$$

$$(-1)! = \frac{0!}{0} = \frac{1}{0} = \infty$$

$$(-1)! = \infty$$

Step 2

Let us assume P(k) to be true

So,

$$(-k)! = (-1)^{k+1} \frac{\infty}{(k-\bar{1})!}$$

Step 3

Therefore, P(k-$\bar{1}$) is also true

So,

$$(-(k-\bar{1}))! = (-1)^{(k-\bar{1})+1} \frac{\infty}{((k-\bar{1})-\bar{1})!}$$

$$(-k+\bar{1})! = (-1)^{(k-\bar{1}+\bar{1})}\frac{\infty}{(k-\bar{1}-\bar{1})!}$$

$$(-k+\bar{1})! = (-\bar{1})^{k}\frac{\infty}{(k-\bar{2})!}$$

Using the factorial property

$$(-n)! = \frac{(-n+\bar{1})!}{(-n+\bar{1})} \qquad \text{or} \qquad (-n+\bar{1})(-n)! = (-n+\bar{1})!$$

$$(-k+\bar{1})(-k)! = (-1)^{k}\frac{\infty}{(k-\bar{2})!}$$

As we know $(-k)! = (-1)^{k+1}\frac{\infty}{(k-\bar{1})!}$

$$(-k+\bar{1})(-1)^{k+1}\frac{\infty}{(k-\bar{1})!} = (-1)^{k}\frac{\infty}{(k-\bar{2})!}$$

$$(-1)(k-\bar{1})(-1)^{k+1}\frac{\infty}{(k-\bar{1})!} = (-1)^{k}\frac{\infty}{(k-\bar{2})!}$$

$$(k-\bar{1})(-1)^{k+2}\frac{\infty}{(k-\bar{1})!} = (-1)^{k}\frac{\infty}{(k-\bar{2})!}$$

$$(-1)^{k+2}\frac{\infty(n-\bar{1})}{(k-\bar{1})!} = (-1)^{k}\frac{\infty}{(k-\bar{2})!}$$

As $(-1)^{k+2} = (-1)^{k}$, we get

$$(-1)^{k}\ \frac{\infty(k-\bar{1})}{(k-\bar{1})!} = (-1)^{k}\frac{\infty}{(k-\bar{2})!}$$

Also, as $\frac{(n-\bar{1})!}{(n-\bar{1})} = (n-\bar{2})!$, we get

$$(-1)^{k}\ \frac{\infty}{(k-\bar{2})!} = (-1)^{k}\frac{\infty}{(k-\bar{2})!} \tag{89}$$

So, we get LHS = RHS, since it is following the dominos effect hance proved that the equation is true for all negative integers.

But it is important to understand why this equation should be defined as a valid formula for negative factorial. So, the key facts to consider this as negative factorial are

- The equation for positive factorial is used to derive the equation for negative factorial and so it is directly related.
- Also, the way the gamma function matches the value with the equation for positive factorials in the same way, the gamma function matches the value for negative factorials. Also, the gamma function is used to derive the value of half factorial which is commonly used in mathematics.

In the below table 1, we will evaluate the results that are obtained using the gamma function as well as using the above mentioned algebraic equation for negative factorials.

The gamma function is expressed as

$$\Gamma(n) = \int_0^{\infty} e^{-x} x^{n-\bar{1}}\, dx$$

And

$$n! = \Gamma(n+1)$$

But as output values are infinitely high and calculation tools have their limitation as they are currently advanced till the real and complex number system and not till the universal number system, so the value of infinity will be replaced with a very large value such that the infinity value is very large real number value compared to other value in the equation. And the infinity value will be accordingly implemented for all other values such as zero and zero square.

The replacement value for infinity and zero are

- Trial 1 : ∞ = 10^5
- Trial 2 : ∞ = 10^6
- Trial 3 : ∞ = 10^7

To evaluate the gamma function, the wolfram online calculation tool is used and the input formula used are

- Trial 1 : integral_0^(10^5) x^((n + 1/10^5)−1) e^(−x)dx
- Trial 2 : integral_0^(10^6) x^((n + 1/10^6)−1) e^(−x)dx
- Trial 3 : integral_0^(10^7) x^((n + 1/10^7)−1) e^(−x)dx

Also, the positive and negative sign of all the outcome value does get reversed when the direction of approaches to n is reversed from RHS to LHS, and so the input equations are

- Trial 1 : integral_0^(10^5) x^((n−1/10^5)−1) e^(−x)dx
- Trial 2 : integral_0^(10^6) x^((n−1/10^6)−1) e^(−x)dx

- Trial 3 : integral_0^(10^7) x^((n−1/10^7)−1) e^(−x)dx

n	$(n-\bar{1})$	(-n)!	$\Gamma(n-\bar{1})$	(-n)!	$\Gamma(n-\bar{1})$	(-n)!	$\Gamma(n-\bar{1})$
		∞ = 100000		∞ = 1000000		∞ = 10000000	
-1	0	-100000	-100001	-1000000	-1000000	-10000000	-10000000
-2	-1	100000	99999.6	1000000	1000000	10000000	10000000
-3	-2	-50000	-49999.5	-500000	-500000	-5000000	-5000000
-4	-3	16666.6	16666.5	166666.6	166666	1666666.7	1666667
-5	-4	-4166.6	-4166.6	-41666.6	-41666.6	-416666.7	-416667
-6	-5	833.333	833.319	8333.33	8333.32	83333.33	83333.33
-10	-9	0.275573	0.275567	2.75573	2.75573	27.5573	27.5573

Table 1: Empirical value for negative factorial using simple Algebraic formulation and using gamma function

So, the gamma function also acts as additional proof and validation for the equation of negative factorials.

Note: While using an online calculator, the input value for integral zero to infinity, zero is taken as 0 and not inverse 1/10^7 as using 1/10^7 value gives the solution with an imaginary magnitude component in it. So, zero is used and solutions are obtained as mentioned above table. It may be due to some zero components are not being handled properly.

10. Basic Algebra: Division by Zero

10.1 Operation in real number system

Let us consider

$a = b$

Multiplying both sides by a

$a^2 = ab$

Adding both sides by a^2

$a^2 + a^2 = ab + a^2$

$2a^2 = a^2 + ab$

Subtracting $2ab$ both sides

$$2a^2 - 2ab = a^2 + ab - 2ab$$

$$2a^2 - 2ab = a^2 + ab - ab - ab$$

As $+ab - ab = 0$

$$2a^2 - 2ab = a^2 - ab$$

$$2a^2 - 2ab = a^2 - ab$$

$$2\left(a^2 - ab\right) = a^2 - ab$$

Dividing the entire equation by $a^2 - ab$

$$\frac{2\left(a^2 - ab\right)}{a^2 - ab} = \frac{a^2 - ab}{a^2 - ab}$$

$$2 = 1$$

But if we do same operation in universal number system by following the rules of universal number system, we get the correct solution.

Let us consider

$$a = b$$

Multiplying both sides by a

$$a^2 = ab$$

Adding both sides by a^2

$$a^2 + a^2 = ab + a^2$$

$$2a^2 = a^2 + ab$$

Subtracting $2ab$ both sides

$$2a^2 - 2ab = a^2 + ab - 2ab$$

$$2a^2 - 2ab = a^2 + ab - ab - ab$$

$$2a^2 - 2ab = a^2 - ab + \mathbf{ab} - \mathbf{ab}$$

As we know $a^2 = ab$

$$2a^2 - 2ab = a^2 - ab + \mathbf{a^2} - \mathbf{ab}$$

$$2\left(a^2-ab\right)=a^2-ab+\mathbf{a^2}\mathbf{-ab}$$

Dividing the entire equation by a^2-ab

$$\frac{2\left(a^2-ab\right)}{a^2-ab}=\frac{a^2-ab}{a^2-ab}+\frac{\mathbf{a^2}\mathbf{-ab}}{a^2-ab}$$

$$2=1+1$$

$$2=2$$

10.2 Inaccuracy in Algebraic Formula

There are many formulas in basic algebra where the accuracy of the formula is compromised at the universal number system level. However, it is correct in the real number system. So such formulas hold the potential to create an invalid solution if the operational range is more than one dimension. In other words, it can create an invalid solution while dealing with zero and infinities.

One of the commonly used formulas is

$$\left(a+b\right)\left(a-b\right)=a^2-b^2$$

But if we see the entire expansion of $\left(a+b\right)\left(a-b\right)$

We get

$$\left(a+b\right)\left(a-b\right)=a^2-ab+ab-b^2$$

Which can be further reduced to

$$\left(a+b\right)\left(a-b\right)=a^2-b^2+ab(0) \tag{90}$$

And we see that $ab(0)$ is ignored which holds the potential to create the instability of mathematical equations.

10.3 Indeterminate forms

$$1^{\infty}=1 \tag{91}$$

$$1^{1/(1-1)}=1$$

$$1=1^{(1-1)}$$

$$1=1^{(0)} \tag{92}$$

In the real number system, zero magnitudes are not handled equally which creates problems to form an indeterminate forms.

It treats

$$1 + 0 = 1 \qquad \textit{(Real numbers system)}$$

But

$$1 + 0 \neq 1 \qquad \textit{(Universal numbers system)}$$

So, for the real number system

$$1^{\infty} = (1 + 0)^{\infty} \qquad \textit{(Real numbers system)}$$

Also, real number treats all or any magnitude of infinity as a single number or concept known as infinity

$$x \times \infty = \infty : x \in \mathbb{R} \qquad \textit{(Real numbers system)}$$

$$x \times \infty \neq \infty : x \in \mathbb{R} \qquad \textit{(Universal numbers system)}$$

And these two inaccuracies of real numbers lead to an indeterminate solution for the problem 1^{∞} as this problem is also treated as

$$1^{\infty} = (1 + 0)^{x\infty} \qquad \textit{(Real numbers system)}$$

$$1^{\infty} \neq (1 + 0)^{x\infty} \qquad \textit{(Universal numbers system)}$$

And, so the above equation has multiple solutions for the various value of x.

So, in the universal number system,

$$1^{\infty} = 1$$

And this is the unique solution to this problem.

And if we add 0 in the numerator then we get

$$(1 + 0)^{\infty} = e$$

Also,

$$(1 + 0)^{x\infty} = e^{x}$$

So, in the universal number system, 1^{∞} and $(1 + 0)^{\infty}$ are not the same and these equations have unique solutions.

10.4 Problems

Show that $x - x$ operation is the dimension resetter that brings value to one dimension cycle backward.

Solve (a): $0 - 0$, (b): $1 - 1$, (c): $\infty - \infty$ & (d): $\infty^2 - \infty^2$.

Solution

(a): $0 - 0 = 0(1-1) = 0(0) = 0^2$

(b): $1 - 1 = 0$

(c): $\infty - \infty = \infty(1-1) = \infty(0) = \infty(0) = 1$

(d): $\infty^2 - \infty^2 . = \infty^2(1-1) = \infty^2(0) = \infty \times \infty(0) = \infty \times 1 = \infty$

Evaluate 0^0.
Also, evaluate the below statements from the real number system that are

- Anything to the power zero is always one
- Zero to the power anything is always zero

Considering both statements to be true but contradicts for 0^0, find which statement is accurately correct or both are correct or both are incorrect?

Solution:

(a)

$$0^0 = 0^{(1-1)} = \frac{0}{0} = 1$$

(b)

$$x^0 = x^{(1-1)} = \frac{x}{x} = 1$$

(c)

$$0^x = 0^x = (1-1)^x$$

- If x is ∞

$$0^\infty = \left(\frac{1}{\infty}\right)^\infty = \frac{1}{\infty^\infty}$$

- If x is 1

$$0^1 = 0$$

- If x is 0

$$0^0 = 0^{(1-1)} = \frac{0}{0} = 1$$

- If x is 0^2

$$0^{0^2} = 0^{0\times 0} = \left(0^0\right)^0$$

Using equation (18), we get

$$\left(0^0\right)^0 = 1 + \frac{\ln(0^0)}{\infty} = 1 + \frac{0\ln(0)}{\infty} = 1 + 0\times 0\ln(0) \approx 1$$

So in the real number system, any number between zero to infinity raised to power zero is one but zero to power anything varies. So, zero to the power of any number from the infinity dimension and above will be approximately zero or a smaller magnitude than zero. Zero to the power of any number from the standard dimension will be approximately zero. Zero to the power of any number from zero dimension and below will be approximately one.

11. Single Solution Formulates

Below are mathematical equations that may form an indeterminate form in a real number system. However, in the universal number system, it has a specific outcome for specific input.

$$\left(1+\frac{1}{\infty}\right)^{\infty} = \left(1+\frac{1}{k\infty}\right)^{k\infty} = e \tag{89}$$

$$\left(1+\frac{k}{\infty}\right)^{\infty} = \left(1+\frac{1}{\infty}\right)^{k\infty} = e^k \tag{90}$$

$$\infty\left(e^{\frac{k}{\infty}}-1\right) = k \tag{91}$$

$$\infty^{-\frac{k}{\ln(0)}} = e^k \tag{92}$$

$$\infty^{\frac{k}{\ln(0)}} = \frac{1}{e^k} \tag{93}$$

$$\infty^{\frac{\ln(k)}{\ln(0)}} = \frac{1}{k} \tag{94}$$

$$\infty^{-\frac{\ln(k)}{\ln(0)}} = k \tag{95}$$

$$\infty \sqrt[\infty]{e^k} - \infty = k \tag{96}$$

Part III – Future Research and Development

12. Imaginary Numbers on Real number plain

The fundamentals of imaginary numbers are stably established based on what was already known at that time. But understanding universal numbers has created more understanding of the fundamentals of entre mathematics giving the chance to see the various mathematical concept from a very new perspective.

As we know, the roots of negative real numbers are defined by an imaginary number $\boldsymbol{i}$ that is on a completely different plane. This means we cannot directly relate these numbers to real numbers. Eg: We can mention that infinity is more than one and one is more than zero, or we can approximate infinity plus one as infinity but we cannot say anything if two is greater or smaller than $3\boldsymbol{i}$. We cannot do such comparisons as it is not that closely related to real numbers but at closest these imaginary number can get transformed into a real number in certain operation such as raising the power by square. And this is how closest these imaginary numbers are currently defined and related to real numbers.

12.1 Solution of negative real roots on universal number plain

More closer the fundamental relationship of negative real roots will be to the real numbers itself, the better mathematical understanding will exist for negative real roots. And this will increase solving capabilities of entire mathematics in the domain of roots and exponents. And understanding the universal number, it is understood that there exists a possibility to define the negative real roots in more closer order in real number plain itself.

Solution of negative real roots on Imaginary numbers plain

Currently, the solution to all negative real number roots is defined on an entirely different plain which is an imaginary number plain. This all starts from the square of a number. A number that can be multiplied to itself and end up as a negative number. However, taking every possible number from our existing mathematics, we end up having 2 different types of numbers, one is a positive number, and another in negative but multiplying a negative number to itself turns out to be a positive number and also multiplying a positive number to itself turns out to be positive numbers as given below.

$$(-a) \times (-a) = +b$$

$$(+a) \times (+a) = +b$$

So, we don't have any other number on the entire real number plain that is different from positive or negative numbers nor this gives the result that we are looking for. This concludes, there must be another number plain that has such property that if it is multiplied by itself gives the negative real number. And this new plain is known as the imaginary number plain. And we have all the solution that is required to deal with roots.

Solution of negative real roots without imaginary number line.

The introduction to imaginary number plain happened due non-existing of a number with a property that can give a negative solution if it is multiplied to itself. However, understanding the universal numbers, excluding the imaginary number, we get two other loopholes where this can be handled and those are:

- It is not the number rather it's the signs that are derived in a certain way that it results in outputs for various combinations, and based on different philosophies and perspectives behind this operation, these solutions will change for different combinations of positive and negative sign operations.
- It is the zero where these signs play the trick. Zero is the place where both positive and negative combine and both of these sign disappears. So zero is the place where either both the sign exists or no sign exists.

So, now we have the above two options where a number can behave in a such property that we can yield a solution for negative real roots. So, out of the above two loopholes, we can neglect the first point and consider only the second point as the source for the second point is the first point itself. Also, the sign cannot enter into the equation by itself, it needs some number to be accompanied with and so it is the second point to be best fit for consideration.

So, it is zero which is the number that has the property to be both positive and negative without clearly getting distinguished whether it is positive or negative. However, we have earlier defined in section 7.2 and 7.3 that zero always yields a positive zero by default. But that is the restriction we have created to solve the problem easily without running into a complicated and double solution, where we have to choose the right one out of two from double solutions. For example:

$$a - a = \pm a\,(0) \tag{97}$$

In this example, whether we mark this RHS of equation (97) as positive or negative, both cannot be proven wrong, rather both can be proven right.

Let us define zero in the given equation (98).

$$(1-1) = 0 = a \tag{98}$$

And with the current knowledge, we can prove the value of (1-1) as positive or negative zero and this sign can keep changing infinite time without breaking any stability of the equation. So zero is holding a dual nature property to be positive and negative. So zero can be expressed as

$$(1-1) = +a \quad || \quad (1-1) = -a \tag{99}$$

But note that the above $(1-1) = +a$ can be still converted into $(1-1) = -a$. And $(1-1) = -a$ into $(1-1) = +a$, this means this form can switch its sign as many as times it wants and if we want to know only one sign of the zero then we have to change its form to another non-zero expression.

Property of Zero

- The sign of the zero is either both or nan.
- Zero is the probability function that can end up at a single solution but until it ends, it remains the possibility be anyone of both solutions when converted to another expression.
- If we have to consider only one sign of the zero, we can consider any sign of zero but it will not be permanent until it is converted into a non-zero expression.

And this is an exact property that imaginary numbers are created with. So, in other words, we can use this zero to solve the solution for negative real roots.

$$x^2 = -a$$

x^2 can be also expressed as

$$x = \sqrt{-a}$$

$$x = \sqrt{x \times x}\sqrt{-1}$$

$$x = x\sqrt{(\infty \times 0)(\infty \times 0)}$$

$$x = x \times 0 \times \infty$$

Therefore the solution to the negative real root is $x = x \times 0 \times \infty$. And this solution is on the same plane as that of the real number line. So, to free the equation from being positive or negative, we can finalize the solution by solving it $0 \times \infty = \pm 1$.

Therefore

$$x = \pm x$$

So, this has two solutions just like the solution for positive root. And an incorrect solution from two solutions will be ruled out based on further solutions.

So the general solution for negative real roots is on the same plane as real number line but this solution will exist in a universal number system and not in a real number system as the concepts of the real number system will not support this solution.

$$x^2 = \sqrt{-a} \tag{100}$$

$$x^2 = \sqrt{(x \times \infty \times 0)(x \times \infty \times 0)} \tag{101}$$

Based on an understanding of the above mathematical expressions, we can express the equations on universal numbers on the same plain as given in figure 8.

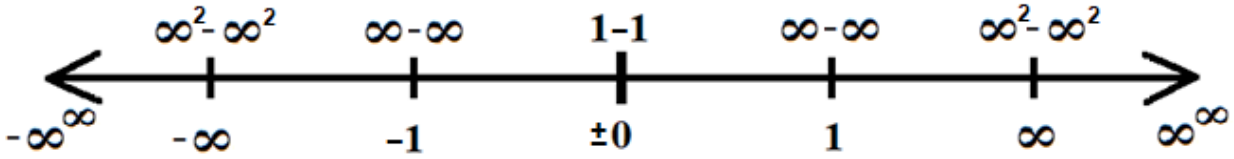


Figure 8: Probability expression equations on universal number line

But note that the equations are probability equations and not static expressions to have a single solution. And this is why a single expression is mapped at two different points as it can lead to any one of those points. For example, the solution to $\infty - \infty$ can be both positive one or negative one.

Part IV – Results and Conclusions

12. Results and conclusion

Subtraction rule

The zero magnitude of the zero dimension also must be balanced in the equation which adds to the accuracy of the equation in the universal number system. So, when any subtraction is conducted, the magnitude of the zero must be equally balanced on both sides of the equation too.

BBUM

Building block unit of a magnitude (BBUM): The BBUM is the magnitude equivalent to one full dimension magnitude back from the given magnitude. BBUM is a relative magnitude unit that varies for all magnitudes. This short form must be pronounced as boom.

Example:

BBUM of 1 is 0

BBUM of ∞ is 1

BBUM of ∞^2 is ∞

BBUM of 3 is $3(0)$

BBUM of 1024 is $1024(0)$

BBUM of $5(0)$ is $5(0^2)$

Remarks (7): In any equation system, it takes more magnitude to come back than it took to reach there.

Remark (8): It takes something more to reverse back to its initial position.

Once something goes to some point with magnitude x, it cannot come back to its initial place with the same magnitude x, it needs a little more magnitude that gets consumed while reversing the direction to come back to its initial position. It costs more. And the cost is one building block unit of that magnitude.

Negative factorial

While using formulas $n(n-1)!=n!$, this works in certain logic rather than a simple pure single mathematical equation.

$$n!=\begin{cases}1 & if\ n=0\\ n(n-1)! & if\ n>0,\ n\in\mathbb{N}\end{cases}$$

From the above fragmented equation, it can be seen that the factorial of zero is one and defined with some logical sense rather than the pure mathematical single equation. And this formula is capable of handling up to whole numbers domain and not beyond. So, a new formula is constructed using $n!=n(n-1)!$ to get value for negative integers too that is expressed as equation (58).

General Conclusion

The real number is well established and it operates at a certain level of accuracy. On the other side, the universal number system is the new system operating at a higher level of accuracy that has a lot more entities and possibilities than the real number system. Its precision is so high that, the real number system won't be able to differentiate the difference between 0.999999….. and 1, in fact, up to a certain extent, the real number system will consider both numbers to be the same, but in a Universal number system, one can fit entire real number system between 0.999999….. and 1.

Playing with the universal number system increases the accuracy at a very super high level which will increase the difficulty level of math operations. Though the difficulty level may be higher than dealing with natural numbers or real numbers, but it is less than dealing with trigonometry or calculus.

Also, this accuracy forces us, not to use certain formulas from our modern mathematics as some formulas in modern mathematics are accurate only till the real number domain and not beyond the real number domain due to rounding off or ignoring the zero terms from the equation. So, such inaccurate formulas at the universal number level cannot be used in this system unless those formulas are rederived while following the accurately proven rules at the universal number system level.

In current use, the universal number should be used only if there is a need to convey the accuracy of values at the universal number level, and it is not required otherwise as its current stage development is very little and it may require R&D frequently whenever the equation gets into a complex stage for which solution will be difficult solve. But this has applications in every part of mathematics and every math user should know the basic concept of the universal numbers

system as it may be used to understand the question itself. For example, the problem 1-1+1-1+-1…… have many solutions because we won't able to frame the question accurately using the real number system and so in such condition, we have to keep the concept of the universal number system in mind and the reader should also able to understand the question accurately.

Real number vs Universal Numbers

The real number system is well developed system and has the best understanding that what the real number system can do and what it cannot do in terms of application and operations. Indeterminant forms and restrictions in the division by zero are some of its limitations. It also has a precise set of restrictions with the condition that what operation will break the stability of the equation and what not, such as any number can go in the denominator but not zero or infinity. And if it goes, the answer will be undefined or infinity (infinity has no precise value understanding in the real number system), and once magnitude is identified as infinity or zero, it can never come back to the standard dimension(real number line). In simple order, the magnitude that goes in zero or infinity dimension will be not in precise control of the real number system, and nor it comes back to standard dimension by any operation.

Also, many functions have function curve points in inverse order where the smallest value of domain(input) such as zero or one yields the infinite range(output). And such functions can be handled in a very controlled order in the universal number system.

Universal number systems have opened a new number line which is the universal number line in which magnitude can travel back and forth in any dimension and can come back to the standard dimension without breaking the stability of equality, and in short, this is what the universal number system can do.

What is infinity in the physical world: maybe it is like water drops that fall continuously. For example: If the ocean is like magnitude one then magnitude zero is like water drop which accounts as a zero ocean. But if that zero falls continuously nonstop and frequently in one place, mathematically it will be taken $x = n \times 0 \times \infty$ where x is the ocean, infinity is because it is continuous, and n determines at what rate it is continuous.

Approximation and Accuracy of the Number System

To make calculations clean, simple, easy, and faster, the equations can be approximated at a certain level. But it will also create the limitation that certain operations cannot be done, or while doing the operation, the overall magnitude of the equation cannot cross a certain magnitude level. And if done, it will give a wrong solution. As approximating the equation includes the ignorance of some smaller or negligible terms which majority of the situation does not create the problem but in some operations even the ignorable magnitude counts which leads to yielding the wrong

solution. However, it is not necessary that every time all equations must be calculated at the highest accuracy level such as maintaining the accuracy at the universal number system level. In fact, for some everyday real problems, accuracy at the real number system level is also not required, the accuracy of natural numbers is enough. So, while solving the problem, we have to take care of what we are approximating and to what domain that solution will work approximately correct, as various operations can push the equation in different domains. So, in short, the operation itself should be avoided if the solution is an approximate solution.

In the Universal number system, the accuracy are followed at 3 levels which if unfollowed is seen as an approximation. Based on the level of these accuracies being maintained or approximation being done, various pros and cons do affect the equation and mathematical problem.

Accuracy Level 1: Basic Level

$$x + m(0) = x + m(0)$$

In this level, the LHS of the equation is not approximated as the RHS of the below equation as the way it is done in real number equations. Rather it is kept as the RHS side of the above equation. As every zero magnitude counts at the zero dimension level.

$$x + m(0) = x$$

Non-following of this accuracy has the following pros and cons.

Pros

Subtraction with the same magnitude is always zero. Also, multiplication by any magnitude with zero is always zero. And these approximation makes the problem much simpler to solve.

Cons

The solution will turn incorrect if it gets divided or multiplied by higher or lower dimension magnitudes. In other words, we cannot divide by zero or multiply by infinity.

The equation can never go in the higher or lower dimension of the magnitude. Magnitudes will get restricted in standard dimensions and hence restricting the boundaries of mathematics itself.

Note: Zero is an invalid solution, if the subtraction magnitude is from a lower or higher dimension as the solution is not approximately close to zero. For example, infinity minus infinity is one, and one is not approximately the same as zero in the standard dimension.

Remark (9): This approximation is followed in the real number system.

Remark (10): This accuracy is fully followed in the Universal number system.

Accuracy Level 2: Standard level

$$a / a = a^0 = 1 + 0 \ln(1)$$

At this accuracy level, anything divided to itself or anything to power zero will be not approximated to one as it is done in a real number system, represented in the below equation. Rather its solution will remain as the RHS of the above equation.

$$a^0 = 1$$

Non-following of this accuracy has the following pros and cons.

Pros
Anything divided to itself or anything to the power zero is one, which makes equations and problems easier to solve.

Cons
The solution will yield the wrong solution if its outcome result is raised(power) by higher or lower dimension magnitudes. In other words, we cannot raise the solution by zero or infinity.

Remark (11): For solving the common problem, this approximation is followed in the universal number system but accuracy needs to be maintained if the power of the equation will be raised by zero or infinity, or while deriving the general mathematical formulas.

Accuracy Level 3: Ultra level

$$1^x = 1^x$$

One to the power anything will remain as mentioned in RHS of the above equation and not approximated to one.

Non-following of this accuracy has the following pros and cons.

Pros:
One to the power anything is one which many problems easier to solve.

Cons:
The solution will yield wrong it is raised by miniature Dimension and supreme Dimension. In other words, we cannot raise the power of the solution by zero square or infinity square.

Remark (11): This approximation is currently rarely followed in the universal number system such as while dealing with log(1) but as the need for accuracy goes higher, this will be regularly followed.

Accuracy Level 4: Probability Accuracy

This is not a fully proven and accepted case. This accuracy will be implemented if the zero is proven to exist in a dual nature which exists as a positive and negative number. This can be also seen as a probability function that holds two solutions within itself. So, to implement, it needs to be fully tested in every scenario to check the stability of existing equality and the concept of mathematics.

Quantum Accuracy

Combined all the above 4 levels will be termed as quantum accuracy. The term quantum is used as this word broadly discusses two major aspects. The first one is the entity to exist in different dimensions such as higher dimension, lower dimension, atomic dimension, subatomic dimension, etc which is similar to the first 3 levels of accuracy. And the second aspect is the probability that an entity exists at two different places or multiple places but there is actually only one entity and generally it remains in probability and not in a known fixed place. And this is similar to the solution of level 4 accuracy about zero to act as a probability function and have multiple solutions in general expression.

Reference

[1] Shah, B. A. (2020, June 15). Universal Numbers - The Universe Of Eternity. https://doi.org/10.35543/osf.io/b76e8

[2] Shah, B. A. (2020, April 4). Special equation from Binomial transform and nth order finite difference. https://doi.org/10.35543/osf.io/xze4u